Relation statistics in a vacuum plane-wave diamond: tidal response, dimension bias and window dependence

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Abstract

Ricci-flatness does not remove the finite-diamond curvature response of the relation fraction. For a four-dimensional constant vacuum plane wave, we evaluate its mean and shared-event covariance by explicit subinterval-volume integrals. With unit null-coordinate span and amplitude A, these are $p_D = 1/10 - 23A^2/75600 + O(A^4)$ and $\zeta_D = 1/100 - 211A^2/4158000 + O(A^4)$. The mean, interpreted through the flat Myrheim–Meyer relation, gives $d_{\mathrm{pop}} = 4 + 23A^2/6363 + O(A^4)$ although the manifold dimension is four. This supplies a vacuum benchmark for separating tidal curvature from apparent dimension change. A normal-coordinate volume calculation agrees with exact slicing; we also document a coefficient mismatch with a displayed literature formula. Fixed coordinate boxes provide controlled contrasts: anisotropy permits linear response, and a square box has the opposite quadratic sign to the diamond. The standard U-statistic variance connects the calculated moments to finite-sample detection forecasts. A previously frozen validation on 51,200 independent fixed-size orders supported all eight forecasts at the prescribed ± 10 percentage-point interval resolution, with flat false-positive upper bounds below 5%. The diamond confirmation concerns a weak-amplitude control, not a measured detection transition. Neither a general Weyl estimator nor an obstruction for the full causal order is established.

1 Question and relation to previous work

Finite causal orders contain statistical information about the Lorentzian regions from which they are sampled. Bombelli's finite-n distributions already provide a framework for comparing geometries in this way [1]. Roy, Sinha and Surya [2] express chain counts through nested interval volumes, calculate leading Ricci corrections, and identify shared-event fluctuations. Kambor and Nomaan [3] study the resulting estimators numerically. We use these established ingredients. Our question is the explicit next response in a tractable vacuum geometry: what remains in the mean and noise when the Ricci terms vanish, and what does that response imply for interpreting one finite order?

Window choice matters because two operations must be distinguished. Keeping a coordinate box fixed changes the causal relation at fixed volume measure. Keeping the tips of an Alexandrov interval fixed also changes its boundary and normalization. Work on several definitions of small causal diamonds already demonstrates the importance of the region definition [4]. A volume expansion alone is not the normalized relation probability: the outer domain and the subinterval orientations must also be integrated. We evaluate those integrals exactly as low-dimensional representations in a constant-profile plane wave, including the shared-event covariance.

Recent work proposes Weyl-sensitive Hasse-diagram observables [5] and investigates conditional sampling costs and bulk response channels [6]. The paired flat/curved constructions and strata in the former use different inputs from our test on one order with known sampling protocol.

Neither method is ranked against the other here. Graph observables also discriminate multiple causal-set ensembles [7]. An APS abstract describes plane-wave Weyl tests through geodesic deviation [8]; only its abstract was available for this comparison. We make no claim to the first Weyl detection, an exhaustive priority search, or reconstruction of an arbitrary curvature tensor.

The proposed contribution is the explicit vacuum-diamond mean and covariance, their tidal and dimension-calibration consequences, and a comparison with coordinate windows. The general nested-volume method and the variance identity are not contributions claimed here. A formula-level comparison also finds a specific volume-coefficient disagreement with Eq. (79) of arXiv:1904.01034v2 [4]; Appendix A supplies two coordinate calculations and a check of that reference's intermediate expressions. This is a comparison in one spacetime, not a replacement general formula. The dimension consequence and this comparison were developed after confirmation; the independent numerical validation below retains its original forecasts and verdicts. An earlier exploratory repository programme motivates the work but is not presented as additional new evidence.

2 Geometry, populations and information available

With natural units $c = 1$, consider

$$\mathrm{d}s^2 = 2\mathrm{d}u\mathrm{d}v + A(x^2 - y^2)\mathrm{d}u^2 + \mathrm{d}x^2 + \mathrm{d}y^2. \tag{1}$$

For constant A, the transverse trace vanishes, so $R_{\mu\nu} = 0$; $R_{uxux} = -A$ and $R_{uyuy} = A$ in the usual curvature convention. The metric determinant is -1. Thus nonzero A changes tidal curvature and causal order while the coordinate volume form remains $\mathrm{d}u\mathrm{d}v\mathrm{d}x\mathrm{d}y$. The amplitude is defined in this chosen null frame. With $\Delta u = 1$, its numerical value is the dimensionless parameter $A(\Delta u)^2$; it is not a frame-independent curvature scalar.

We use $0 \le A \le 2$, before the first conjugate separation $\sqrt{A}\Delta u = \pi$. All three boxes have $0 < u < 1$, $-1/2 < v < 1/2$, and transverse full widths (a, b) centred on zero:

$$R_+ : (2, 6), \qquad R_0 : (\sqrt{12}, \sqrt{12}), \qquad R_- : (6, 2). \tag{2}$$

Each has volume 12. The fourth population is the Alexandrov interval D_A between $P = (0, 1/2, 0, 0)$ and $Q = (1, -1/2, 0, 0)$. The axial tip proper time is $\sqrt{2}$ in every arm. Its volume is $\pi/6$ at $A = 0$ and changes with A.

Each realization contains exactly N independent points from the normalized physical volume of its region. This is the fixed-N conditional version of Poisson sprinkling. Independent arms have independent points. The construction specifies a continuum sampling protocol; it does not infer windows or tips from an arbitrary unlabelled order. The statistic receives just the order and N, while its threshold is calibrated to the known region and N. It receives no point correspondence, coordinates, or value of A for the tested realization. Comparisons at fixed N between boxes and diamonds are *not* comparisons at matched physical density.

For the transitive causal relation, let $h_{ij} = 1$ if points i, j are related, and zero otherwise. We study

$$f_N = \frac{1}{\binom{N}{2}} \sum_{i<j} h_{ij}, \qquad p_R(A) = \mathbb{E} f_N. \tag{3}$$

These are relations, not only links in the Hasse diagram. Null boundaries have zero sampling probability, but numerical near-boundary decisions are checked.

3 Exact causal relation

Sort a pair by u and set $s = u_q - u_p$, $d_x = x_q - x_p$, $m_x = (x_p + x_q)/2$, and similarly for y. For $w = \sqrt{A} > 0$, the classical transverse action is

$$\begin{aligned} C_A = {} & w \tanh(ws/2) m_x^2 + \frac{w}{4} \coth(ws/2) d_x^2 \\ & - w \tan(ws/2) m_y^2 + \frac{w}{4} \cot(ws/2) d_y^2. \end{aligned} \tag{4}$$

The pair is causal precisely when $\Delta v + C_A \le 0$. This follows by minimizing $\frac{1}{2}\int (x'^2 + y'^2 + A(x^2 - y^2))\mathrm{d}u$ with fixed transverse endpoints. The transverse equations are $x'' = Ax$, $y'' = -Ay$; their action is minimizing in the stated pre-conjugate domain. Equation (4) is a half-angle evaluation of that standard solution, not a new causal law.

4 Diamond boundary, mean and covariance

For $0 < t < s$, define

$$B_x(t;s) = \frac{w}{2}[\coth(wt) + \coth(w(s-t))], \quad B_y(t;s) = \frac{w}{2}[\cot(wt) + \cot(w(s-t))], \tag{5}$$

and $J(s) = \int_0^s [B_x(t;s) B_y(t;s)]^{-1/2} \mathrm{d}t$. Both coefficients are positive in the specified domain. Completing the two action quadratics about the classical transverse path shows that any subinterval with u separation s and slack $\ell = -\Delta v - C_A(p,q) > 0$ has volume

$$V(p,q) = \frac{\pi \ell^2}{2} J(s). \tag{6}$$

Indeed the intermediate transverse ellipse has height $\ell - B_x \xi^2 - B_y \eta^2$ in v; its integrated slice is $\pi \ell^2 / (2\sqrt{B_x B_y})$. This argument also explains why fixed coordinate volume form does not imply fixed Alexandrov volume.

In the main diamond write $S(u) = [B_x(u;1) B_y(u;1)]^{-1/2}$. The u marginal is $S(u)/J(1)$ and $V(D_A) = \pi J(1)/2$. At fixed u, the squared elliptical radius has density $2(1 - r^2)$; angle is uniform. For width $L = 1 - r^2$ and $z = (v - v_{\mathrm{low}})/L$, L has density $2L$ and z is independent uniform. The conditional probability that an independent point is related to X is

$$q(X) = \frac{L^2}{J(1)} [J(u)(1-z)^2 + J(1-u) z^2]. \tag{7}$$

The past and future subdiamonds lie inside the main diamond, so their normalized volumes can be added. This is the conditional-volume form of the known chain and overlap construction, rather than a new identity. Integrating L, z gives explicit smooth low-dimensional expressions

$$p_D = \frac{1}{6 J(1)^2} \int_0^1 S(u)[J(u) + J(1-u)] \mathrm{d}u, \tag{8}$$

$$\mathbb{E}q^2 = \frac{1}{J(1)^3} \int_0^1 S(u) \left\{ \frac{J(u)^2 + J(1-u)^2}{15} + \frac{J(u) J(1-u)}{45} \right\} \mathrm{d}u, \tag{9}$$

$$\zeta_D = \mathbb{E}q^2 - p_D^2.$$

At $A = 0$, $J(s) = s^2/3$, $p_D = 1/10$ and $\zeta_D = 1/100$.

Proposition 1 (Diamond response)**.** *For these fixed axial tips,*

$$p_D = \frac{1}{10} - \frac{23}{75600} A^2 + O(A^4), \qquad \zeta_D = \frac{1}{100} - \frac{211}{4158000} A^2 + O(A^4). \tag{10}$$

Proof. Use $\sin z \sinh z/z^2 = 1 - z^4/90 + O(z^8)$ in (5). This gives

$$J(s) = \frac{s^2}{3}\left[1 + \frac{A^2 s^4}{252} + O(A^4)\right],$$
$$S(u) = 2u(1-u)\left[1 + \frac{A^2(1 - u^4 - (1-u)^4)}{180} + O(A^4)\right]. \tag{11}$$

Substitution into (8)–(9) leaves polynomial integrals on $[0,1]$ and yields (10). Analyticity in A^2 follows from the displayed sine/hyperbolic-sine product and its positive denominator; the remainder is therefore justified here independently of parity. □

4.1 Tidal meaning and flat-calibration bias

Along the axial geodesic, choose the unit tangent $U = (\partial_u - \partial_v)/\sqrt{2}$ and its orthogonal spatial frame. The electric Weyl tensor $E_{ij} = C(U, e_i, U, e_j)$ has diagonal entries $(-A/2, A/2, 0)$. For tip proper time $T = \sqrt{2}$, the dimensionless tidal strength is $\kappa = E_{ij}E^{ij}T^4 = 2A^2$. Thus

$$p_D = \frac{1}{10} - \frac{23}{151200}\kappa + O(\kappa^2), \qquad \zeta_D = \frac{1}{100} - \frac{211}{8316000}\kappa + O(\kappa^2). \tag{12}$$

This expression retains the diamond's timelike orientation. A single plane-wave family cannot determine separate coefficients of all Weyl contractions in a general vacuum metric. Restoring an arbitrary null span Δu replaces A in (10) by $A(\Delta u)^2$.

The flat ordering probability in dimension d is

$$F(d) = \frac{\Gamma(d+1)\Gamma(d/2)}{2\Gamma(3d/2)}, \quad F(4) = \frac{1}{10}, \quad F'(4) = -\frac{101}{1200}. \tag{13}$$

This normalization is twice $\langle C_2\rangle/\langle C_1\rangle^2$ in the Poisson convention of Ref. [2]. Let d_{pop} solve $F(d_{\mathrm{pop}}) = p_D(A)$ near four. The inverse-function expansion gives

$$d_{\mathrm{pop}} = 4 + \frac{23}{6363}A^2 + O(A^4) = 4 + \frac{23}{12726}\kappa + O(\kappa^2). \tag{14}$$

This is a curvature bias of the *population calibration*; it is not $\mathbb{E}[F^{-1}(f_N)]$, which has additional nonlinear sampling bias. At the flat point, the leading standard error of that inverse is $240/(101\sqrt{N})$. Equations (12)–(14) give a quantitative vacuum benchmark: a positive apparent dimension shift can arise while the actual dimension remains four. Curvature bias in dimension estimation is already known; the result here is its explicit leading coefficient for this Ricci-flat family. No new Monte Carlo test of this corollary is claimed.

4.2 Volume consistency check

The same slice calculation gives

$$\frac{V(D_A)}{V(D_0)} = 1 + \frac{A^2}{252} + O(A^4). \tag{15}$$

For comparison, the conventions of Ref. [4] give $d = 4$, tip half-length $l = T/2 = 1/\sqrt{2}$, $E^2 = A^2/2$, $H^2 = A^2$ and $D^2 = 2A^2$. Substitution into Eq. (79) of its arXiv v2 PDF yields instead $1 + (43/9450)A^2$ at this order. The coefficient difference is $11/18900$. It cannot be attributed to a Ricci term or a curvature derivative: $R_{ab} = 0$ and $\nabla R_{abcd} = 0$ here. Appendix A independently recovers (15) in normal coordinates. Specializing and reintegrating that reference's Eqs. (74),(76) also gives $1/252$. We report this reproducible disagreement with the displayed final expression; we have not derived a corrected formula for general vacuum spacetimes or obtained an external adjudication of the discrepancy.

Table 1: Coefficients in $p_R(A) = p_0 + c_1 A + c_2 A^2 + o(A^2)$, derived without fitting poset samples. The diamond has the stronger $O(A^4)$ remainder. The equal-volume statement applies to the three boxes only.

Region	p_0	c_1	c_2
R_+	0.02380872630	+0.04051927709	+0.05074242895
R_0	0.02454196871	0	+0.00572728722
R_-	0.02380872630	−0.04051927709	+0.05074242895
D	0.10000000000	0	−0.00030423280

5 Coordinate-window contrasts

In a fixed box, s has density $2(1-s)$ on $(0,1)$. Integrating both independent uniform v coordinates gives $p_R(A) = \mathbb{E}G(C_A)$, where

$$G(C) = \begin{cases} 1, & C \le -1, \\ 1 - (1+C)^2/2, & -1 < C < 0, \\ (1-C)^2/2, & 0 \le C < 1, \\ 0, & C \ge 1. \end{cases} \tag{16}$$

The remaining integral is five-dimensional. It handles negative action as well as positive action; retaining only the positive branch would lose part of the finite-amplitude curved population.

Proposition 2 (Local box coefficients)**.** *For the three boxes above, $p_R(A) = p_0 + c_1 A + c_2 A^2 + o(A^2)$, with the coefficients in Table 1 and the exact expressions in Appendix B. The identity $p_{(a,b)}(A) = p_{(b,a)}(-A)$ holds when negative A is defined by exchanging the transverse axes.*

Proof. Expanding the action yields $C_A = C_0 + AC_1 + A^2 C_2 + O(A^3)$, with

$$\begin{aligned} C_0 &= (d_x^2 + d_y^2)/(2s), \\ C_1 &= s\{(m_x^2 - m_y^2)/2 + (d_x^2 - d_y^2)/24\}, \\ C_2 &= -s^3\{(m_x^2 + m_y^2)/24 + (d_x^2 + d_y^2)/1440\}. \end{aligned} \tag{17}$$

The A-dependent derivatives are bounded on these compact boxes, including as $s \to 0$. Since G' is bounded and Lipschitz, dominated convergence applies to the first and second integrated differences, away from the measure-zero surfaces $C_0 = 0, 1$. Therefore

$$\begin{aligned} c_1 &= \mathbb{E}[-(1-C_0)C_1 \mathbf{1}_{C_0<1}], \\ c_2 &= \mathbb{E}[\{-(1-C_0)C_2 + C_1^2/2\}\mathbf{1}_{C_0<1}]. \end{aligned} \tag{18}$$

Conditional on $|d_x|$, the midpoint is uniform on an interval of width $a - |d_x|$, giving second and fourth moments $(a-|d_x|)^2/12$ and $(a-|d_x|)^4/80$. Because $\min(a,b) > \sqrt{2}$, the full displacement disk $d_x^2 + d_y^2 < 2s$ lies in the support. Polynomial disk integration gives the listed expressions. Axis exchange gives the identity and $c_1(R_0) = 0$. Finally, both terms of the c_2 integrand are nonnegative and their sum is positive on a set of positive measure. □

Evenness alone does not justify an $O(A^4)$ box remainder: G'' changes at zero, and higher differentiability of the integrated expression has not been established here. All finite-A forecasts use the full integral, not a Taylor polynomial evaluated outside its controlled local limit. The diamond and square box have opposite leading quadratic responses. The box coefficient is about 18.8 times larger in magnitude, but their volumes and fixed-N noise also differ. This compares the declared protocols, not all ways of measuring curvature.

6 Finite-sample forecast and numerical checks

For independent volume samples, define $\zeta_R = \mathbb{E}[h_{12}h_{13}] - p_R^2 = \mathrm{Var}(\mathbb{E}[h_{12} \mid X_1])$. Counting overlapping pairs gives the standard U-statistic identity [9]

$$\sigma_R^2(A,N) = \mathrm{Var}(f_N) = \frac{2p_R(1-p_R)}{N(N-1)} + \frac{4(N-2)\zeta_R}{N(N-1)}. \tag{19}$$

Treating the $\binom{N}{2}$ relations as independent observations would omit the second term. Our independent replication unit is the complete order.

For all windows, an additional numerical route evaluates p_R and $\mathbb{E}[h_{12}h_{13}]$ from three point backbones (u,x,y). Conditional relation probabilities are clipped affine functions of the anchor's v. Their product is piecewise quadratic and is integrated exactly between breakpoints. The remaining nine-dimensional integral uses scrambled Sobol points. Final forecasts used 32 scrambles of 2^{19} points per flat/curved region condition, independent of all poset seeds. Approximate Bonferroni Student intervals for the moments were propagated through the two normal tails. The largest resulting probability error allowance was 0.002880. This is numerical error evidence, not directed-rounding certification or an exact QMC coverage theorem.

The threshold for a known region and N is frozen as the actual two numbers

$$L_R, U_R = p_R(0) \mp z_{0.995}\sigma_R(0,N). \tag{20}$$

A realization is classified as curved if $f_N < L_R$ or $f_N > U_R$. The forecast is

$$\widehat{\beta}_R(A,N) = \Phi\left(\frac{L_R - p_R(A)}{\sigma_R(A,N)}\right) + 1 - \Phi\left(\frac{U_R - p_R(A)}{\sigma_R(A,N)}\right). \tag{21}$$

Only the variance is exact; the normal distributional approximation needs validation. No finite-N normal-error theorem is asserted.

The instrument was checked before pilot sampling. Polynomial coefficients agreed with separate 24- and 40-node Gauss integrations to less than 2×10^{-13}. A numerical geodesic boundary-value solution and independent action quadrature on 96 prescribed pairs agreed with the analytic action to maximum scaled discrepancy 1.32×10^{-13}. The vector relation evaluator agreed with the existing scalar predicate on 1,280 prescribed pairs across five amplitudes, plus boundary cases. Before confirmation, sixteen targeted tests covered permutation and axis-exchange invariance, conditional integration, chain counting, variance and interval gates. Two additional artifact-audit tests passed after confirmation. A symbolic reconstruction reproduced the response coefficients without reading poset data. A separate post-confirmation audit checked the metric curvature, normal-coordinate volume and dimension-calibration identities added in this revision. These computational checks are not external mathematical review.

Diamond sampling uses rejection from the exact u slice density, which is maximal at $u = 1/2$, followed by the elliptical radial and uniform-v draws above. The maximum follows from concavity of $\log\sin$ and $\log\sinh$. Tip membership was checked on 4,096 points at each of $A = 0, 1, 2$. Numerical volumes were 0.5235987756, 0.5256934164 and 0.5321871128. Only the QMC integration uses a numerical inverse CDF; comparison of 4,097 and 8,193 grid points provided a separate allowance. Actual confirmation samples do not inherit that discretization. The smooth diamond moments independently agreed at 32 and 64 Gauss nodes and with the QMC uncertainty intervals.

7 Prospective validation and results

A fixed development grid used $A \in \{0, 1/64, 1/16, 1/4, 1, 2\}$. The pilot generated 80 independent orders in each region/amplitude condition at $N = 200, 800$, for 3,840 orders. Its established

Table 2: Independent confirmation, $m = 3200$ per row per arm. All probabilities are percentages. Curved CIs and flat upper bounds belong to the common 16-interval, 95% simultaneous sampling family. All eight prediction-error intervals also passed after numerical widening. A star marks the separate 80% detection criterion.

Region	N	A	Pred.	Obs.	Curved CI	Flat upper
R_+	300	0.125	44.84	43.53	[40.94,46.15]	1.98
R_+	1200	0.125	98.24	98.56	[97.83,99.11]	1.72*
R_0	300	$\sqrt{2}$	66.96	66.03	[63.51,68.49]	1.72
R_0	1200	$\sqrt{2}$	99.68	99.84	[99.51,99.97]	1.64*
R_-	300	0.125	20.98	21.13	[19.03,23.33]	1.40
R_-	1200	0.125	91.91	91.56	[90.01,92.95]	1.56*
D	300	0.015625	1.00	1.03	[0.58,1.68]	1.48
D	1200	0.015625	1.00	1.09	[0.63,1.75]	1.52

$R_+, A = 1$ positive control had positive 95% difference intervals at both sizes. The pilot was checked for unexplained mean, variance and tail discrepancies. Compatibility of its small samples with planning assumptions was not treated as proof of those assumptions.

For each region, the first positive-amplitude grid interval in which the $N = 300$ forecast crossed 50% supplied its geometric-mean confirmation amplitude. This selected $A = 1/8$ for R_+, R_- and $A = \sqrt{2}$ for R_0. No crossing occurred in the diamond grid. The prescribed boundary rule selected $A = 1/64$, the nearer positive boundary in forecast probability. Consequently the diamond confirmation is a weak-signal control; it is not a strong-amplitude test. Its A was in the pilot, while both confirmation N values and all confirmation seeds were new. The box amplitudes were also new.

At $N = 300, 1200$, each of eight curved conditions received its own independent flat arm. The 16 primary binomial intervals use two-sided Clopper–Pearson construction with error $0.05/16$ per interval, providing at least 95% simultaneous sampling coverage for the fixed forecasts. Every flat interval's upper endpoint must be at most 0.05. Every curved interval, after subtracting the frozen prediction and expanding by its numerical allowance, must be wholly inside $[-0.10, 0.10]$. A separate detection label requires the curved lower endpoint to exceed 0.8 and its flat arm to pass. Low detection can therefore be accurately predicted without satisfying the detection label.

Before pilot data, power scenarios allowed true flat error in $\{0.01, 0.02, 0.03\}$ and absolute forecast bias in $\{0, 0.02, 0.04\}$. Exact binomial acceptance probabilities and a union bound, with numerical allowances included, selected $m = 3200$ per arm/condition from a fixed ladder. The worst declared scenario had joint-power lower bound 0.9818, exceeding the 90% target. This is conditional design power, not a guarantee when the true populations violate those scenarios. Protocols, numerical forecasts, thresholds, seed namespaces and sample sizes were committed before confirmation. No holdout refitting or optional sample-size extension occurred.

All 16 gates passed (Table 2). The maximum point discrepancy was 1.31 percentage points. No uncertain pair required precision escalation in the confirmation data. Absence of such cases in this finite sample does not prove floating-point correctness on every admissible pair: the guard is an arithmetic model, and near-guard cases would use the legacy fallback or stop. An independent audit recomputed counts, intervals and verdicts from the saved raw statistics and checked their hashes and pre-data forecast commit.

The run used 4,165.05 summed worker CPU seconds and 551.10 wall seconds with eight workers on the local Windows host. Development grid, pilot and forecast calculation used 248.31 summed worker CPU seconds, excluding software authoring, initial instrument checks, process startup and final analysis. The priced confirmation budget with a factor-two allowance was 1.486 CPU-hours; the observed 1.157 CPU-hours remained within it and the 48 CPU-hour cap.

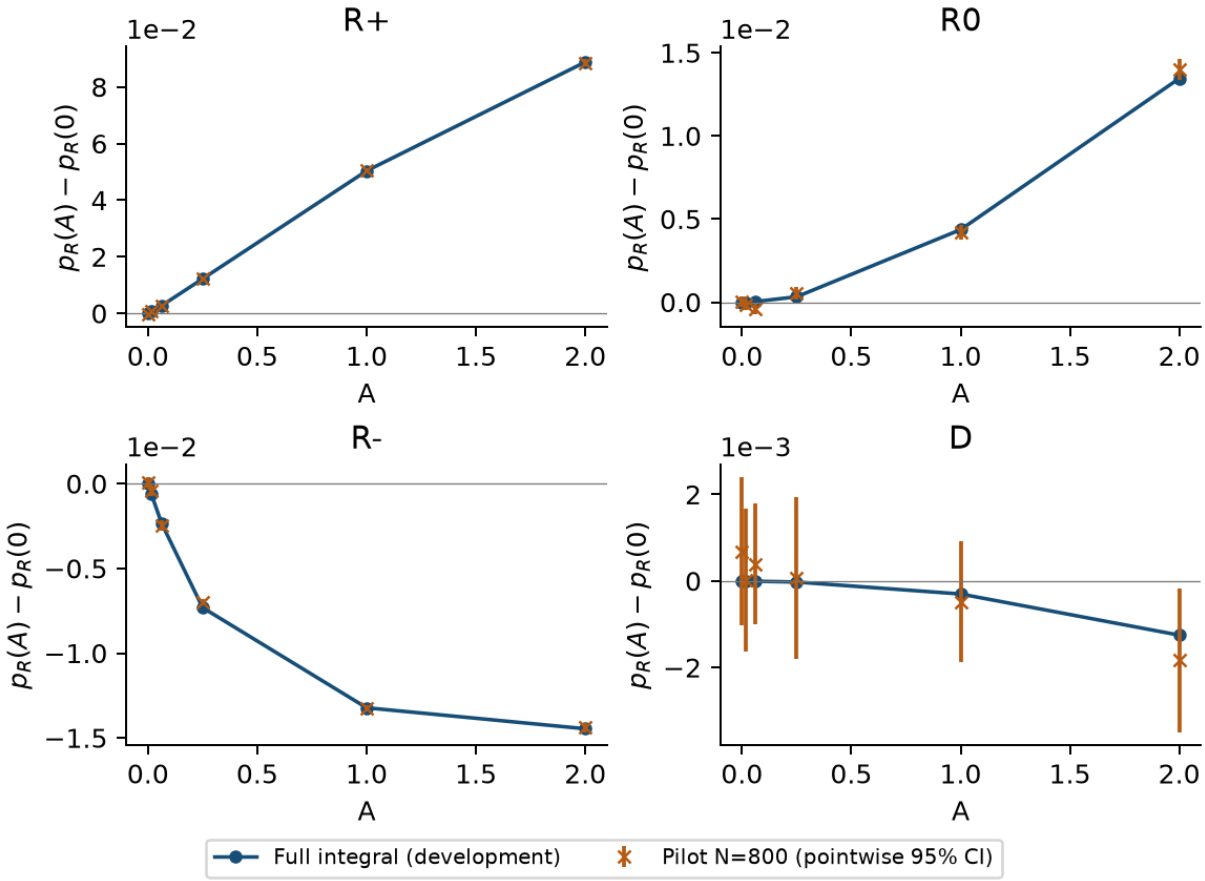


Figure 1: Mean response at common development amplitudes. Lines connect quadrature evaluations and are visual guides. Crosses are pilot means with pointwise 95% intervals at $N = 800$ (80 orders per point). Different vertical scales are explicit. The noisy diamond pilot does not itself determine the sign of its weak response; that sign follows from the analytic calculation.

7.1 Secondary comparisons

The raw data also retain the three-chain fraction $t_{3,N} = \sum_j d_j^- d_j^+ / \binom{N}{3}$. These counts use the transitive relation. Descriptive signed AUCs were oriented using the nearest development pilot condition, never the confirmation labels. At $N = 300$, the respective $(f_N, t_{3,N})$ AUCs were $(0.9426, 0.9437)$ in R_+, $(0.9673, 0.9990)$ in R_0, and $(0.9157, 0.8694)$ in R_-. Thus the extra statistic did not uniformly improve empirical ranking. This post-confirmation comparison has no simultaneous advantage claim and does not replace the primary f_N gate. Both statistics were near chance in the weak diamond condition; noisy pilot orientation there varied with sample size.

Reusing the frozen R_+ threshold without region recalibration gave 100% flat false positives in both diamond samples. The flat means already explain this failure (0.02381 versus 0.1); it is not evidence that curvature itself caused the transfer failure. In R_0 at $N = 1200$, transferred flat error was 3.25%, versus 1.00% under its own calibration. Differences between curved regions additionally involve different selected A and cannot isolate shape alone. One balanced random-label diagnostic per condition gave AUCs 0.487–0.509. It is a diagnostic, not a new hypothesis test.

On a separate $R_0, A = 1$ timing sample, nine warm repeats gave median additional three-chain aggregation times of 0.094 ms at $N = 300$ and 1.319 ms at $N = 1200$, after the relation matrix already existed. Matrix construction took 1.948 ms and 54.258 ms respectively. This small host-specific benchmark measures the incremental count, not a portable hardware performance bound.

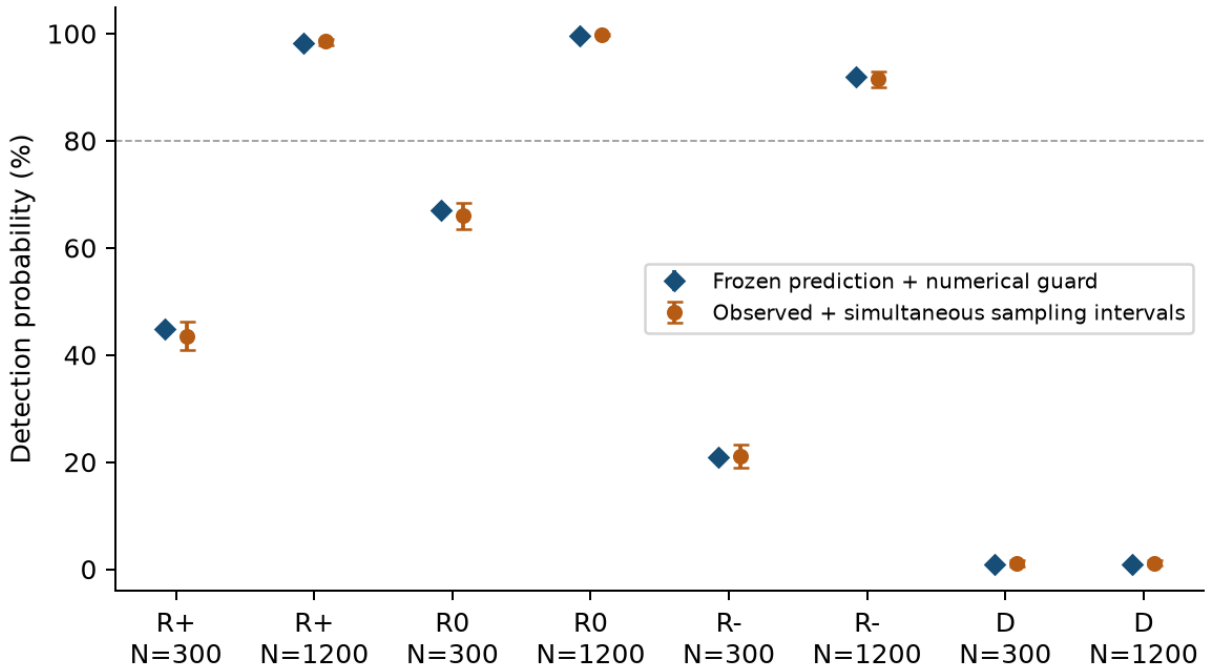


Figure 2: Frozen forecast versus observed detection. Blue guards represent approximate numerical integration uncertainty; orange intervals represent simultaneous sampling uncertainty as specified in Table 2. Amplitudes differ by region. The 80% line is a separate detectability criterion.

8 Interpretation and limits

The diamond calculation identifies an explicit vacuum response beyond the leading Ricci correction used in small-diamond chain formulae. Its practical use is as a benchmark for mean, noise and flat-dimension calibration, with a known geometry and orientation. The volume comparison in Section 4.2 is also a testable consistency question for higher-order diamond expansions. Neither result establishes a universal curvature reconstruction procedure.

The quantitative distinction is between the same order statistic under different acquisition protocols. In the anisotropic boxes, the two transverse tidal contributions are weighted differently and leave opposite nonzero linear coefficients. Equal widths cancel that term. In the moving diamond, the boundary and its normalization enter the response along with the relation; the resulting quadratic coefficient has the opposite sign.

When $\zeta_R(0) > 0$, equation (19) gives standard deviation asymptotic to $2\sqrt{\zeta_R(0)/N}$. A nonzero local mean response cA^k therefore suggests A of order $N^{-1/(2k)}$ at fixed standardized signal. This is a standard conditional asymptotic consequence, not a newly measured scaling law. Two confirmation sizes and moderate box amplitudes do not establish its asymptotic exponent. Nor can the diamond's weak-amplitude result exclude information in higher chains or the full order, establish non-detectability at $A = 2$, or prove a universal lower bound on sample complexity.

The successful forecasts support the declared ten-percentage-point resolution at eight selected conditions. They do not certify exact 1% tail calibration: the formal flat upper bound was 5%. The numerical guards are approximate; the exact simultaneous sampling statement concerns the frozen numerical forecast values. We do not combine these into an exact 95% joint guarantee over an ideal continuum forecast. Extension beyond the pre-conjugate plane wave, to other windows, or to inference of unknown window geometry requires new derivations and independent validation.

Data, code and declarations

The companion source, all 51,200 raw statistic triples, development outputs, protocols, frozen forecasts, validation audit and figure generator accompany this local review manuscript. The project repository is https://github.com/lpaiu-cs/causal-spacetime; the new study is on the local `codex/observation-window-response` branch. A public archive of this study has not yet been deposited. The companion README records producer commits, hashes, software versions and reproduction commands. Earlier frozen results and the earlier submission have not been overwritten.

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A Normal-coordinate volume cross-check

This derivation is independent of the conditional-degree calculation and was added after the frozen campaign. At the axial midpoint use normal coordinates (t, z, X, Y), put $s = (t+z)/\sqrt{2}$, and set $w = \sqrt{A}$. The exponential map obtained from the geodesic equations is

$$\begin{aligned} u &= u_O + s, \qquad x = X\frac{\sinh(ws)}{ws}, \qquad y = Y\frac{\sin(ws)}{ws}, \\ v &= \frac{-t+z}{\sqrt{2}} - \frac{X^2}{2s}\left(\frac{\sinh(ws)\cosh(ws)}{ws} - 1\right) \\ &\quad - \frac{Y^2}{2s}\left(\frac{\sin(ws)\cos(ws)}{ws} - 1\right). \end{aligned} \tag{22}$$

Continuous limits are understood at $s = 0$. The absolute Jacobian is

$$\sqrt{-g_{\mathrm{RNC}}} = \frac{\sinh(ws)\sin(ws)}{(ws)^2} = 1 - \frac{A^2(t+z)^4}{360} + O(A^4 s^8). \tag{23}$$

The tips have normal times $\pm l$. For the upper cone let $r_0 = l - t$, $(X, Y, z) = r(n_x, n_y, n_z)$ and $d_n = n_x^2 - n_y^2$. Substituting (22) in the exact null condition and solving gives

$$\begin{aligned} r &= r_0(1 + Ab_1 + A^2 b_2 + O(A^3)), \\ b_1 &= -\frac{l^2 d_n}{12}, \\ b_2 &= \frac{l^4 d_n^2}{96} + \frac{l^2(1-n_z^2)}{360}[(t + r_0 n_z)^2 - 4l(t + r_0 n_z) + l^2]. \end{aligned} \tag{24}$$

The lower cone gives the same integrated contribution. Its moving join with the upper cone does not change the A^2 coefficient: the first-order time shift is zero and the unperturbed densities agree at the join. Using normalized angular averages $\langle d_n^2 \rangle = 4/15$, $\langle 1 - n_z^2 \rangle = 2/3$ and $\langle (1 - n_z^2) n_z^2 \rangle = 2/15$, one obtains

$$[A^2]V = 8\pi \int_0^l \left[r_0^3 \langle b_2 + b_1^2 \rangle - \frac{1}{360}\left(\frac{t^4 r_0^3}{3} + \frac{2t^2 r_0^5}{5} + \frac{r_0^7}{35}\right)\right] \mathrm{d}t = \frac{2\pi l^8}{189}. \tag{25}$$

Here $[A^2]$ denotes the Taylor coefficient. Since $V_0 = 2\pi l^4/3$, the relative correction is $A^2 l^4/63$, giving $A^2/252$ for $l = 1/\sqrt{2}$. The overall volume is even and analytic in A, as is also manifest in the exact slice formula.

The companion `check_diamond_geometry.py` differentiates the original metric to verify Ricci-flatness, covariantly constant curvature, the tetrad contractions, the exponential-map Jacobian and the null residual through A^2. It separately specializes Ref. [4], Eqs. (74),(76), performs their angular and radial integrals, and obtains the same coefficient. This agreement localizes the comparison to the displayed final Eq. (79), without proposing a general correction. A 60-digit evaluation of the exact slice at $A = 0.1, 0.05, 0.025, 0.0125$ gives $(V/V_0 - 1)/A^2$ respectively as 0.0039685733, 0.0039683338, 0.0039682739 and 0.0039682590, approaching $1/252 = 0.0039682540$. These are deterministic arithmetic checks, not additional statistical evidence.

B Exact box coefficient evaluation

For nonnegative integers i, j and the powers k used here, set $h = (i + j + 2)/2$, $t = k + h$. Integration over the full displacement disk gives

$$\int_0^1 2(1-s)s^k \int_{d_x^2+d_y^2<2s} |d_x|^i |d_y|^j \mathrm{d}d_x \mathrm{d}d_y \mathrm{d}s = \frac{4B((i+1)/2, (j+1)/2)2^h}{(i+j+2)(t+1)(t+2)}. \tag{26}$$

To evaluate (18), multiply the midpoint-averaged polynomial by $(a-|d_x|)(b-|d_y|)/(a^2b^2)$ before using (26). For $(a, b) = (2, 6)$ this yields

$$\begin{aligned} p_0 &= \frac{\pi}{108} + \frac{1}{5184} - \frac{128\sqrt{2}}{33075}, \\ c_1 &= \frac{\pi}{54} + \frac{1}{540} - \frac{352\sqrt{2}}{25515}, \\ c_2 &= \frac{1613\pi}{48600} + \frac{16733}{2177280} - \frac{1316008\sqrt{2}}{30405375}. \end{aligned} \tag{27}$$

For R_- only the sign of c_1 changes. For the square box,

$$\begin{aligned} p_0 &= \frac{\pi}{108} + \frac{1}{5184} - \frac{64\sqrt{6}}{33075}, \\ c_1 &= 0, \qquad c_2 = \frac{257\pi}{48600} + \frac{73}{435456} - \frac{137204\sqrt{6}}{30405375}. \end{aligned} \tag{28}$$

These are exact coefficients in the stated local expansion; the finite-amplitude mean is evaluated with (4) and (16). This distinction is essential in the square-box confirmation at $A = \sqrt{2}$.

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